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November 2024

Acute changes in hamstring injury risk factors after a session of high-volume maximal sprinting speed efforts in soccer players. Sports Health, Ahead of Print: 1-12, 2024.

Adapting the percentage intensity method to assess accelerations and decelerations in football (soccer): Moving beyond absolute and arbitrary thresholds. Sports Biomechanics, Ahead of Print: 1-13, 2024.

American football headgear impairs visuomotor drill performance in Division I NCAA football athletes. Journal of Functional Morphology and Kinesiology, 9: 169, 2024.

Associations between brain metabolites measured with MR spectroscopy and head impacts in high school American football athletes. Journal of Magnetic Imaging, Ahead of Print: 1-13, 2024.

Bidirectional link between exercise and the gut microbiota. Exercise and Sport Sciences Reviews, 52(4): 132-144, 2024.

Circadian variation in coaches' decision-making in the National Football League's evening games. Chronobiology International: The Journal of Biological and Medical Rhythm Research, Ahead of Print: 1-8, 2024.

Effects of cannabidiol ingestion on thermoregulatory and inflammatory responses to treadmill exercise in the heat in recreationally active males. Medicine and Science in Sports and Exercise, Ahead of Print: 1-43, 2024.

Imaging and surgical procedures after surface-related collegiate football injuries on artificial turf versus natural grass. The Orthopaedic Journal of Sports Medicine, 12(9): 1-10, 2024.

In-match strategies to mitigate the effect of heat on football (soccer) players' health and performance. British Journal of Sports Medicine, Ahead of Print: 1-2, 2024.

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Is chronic ankle instability associated with contractile thickness of gluteus medius and gluteus maximus during functional movement and exercise? A systematic review and meta-analysis. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-18, 2024.

Limitations of separating athletes into high or low risk groups based on a cut-off – A clinical commentary. International Journal of Sports Physical Therapy, 19(9): 1151-1164, 2024.

Lower extremity injury rates on artificial turf versus natural grass surfaces in the National Football League during the 2021 and 2022 seasons. The Orthopaedic Journal of Sports Medicine, 12(8): 1-6, 2024.

Medical malpractice lawsuits against team physicians in the National Football League increased between 1971 and 2020 but remain low overall. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-20, 2024.

Observation of risk for concussion following 'lowering the helmet to initiate contact' penalties in the NFL. Journal of Orthopaedic Reports, 3: 100347, 2024.

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The effect of sport specialization on injury risk in NCAA athletes: Results from the SAFE Consortium. Clinical Journal of Sports Medicine, Ahead of Print: 1-5, 2024.

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Acute changes in hamstring injury risk factors after a session of high-volume maximal sprinting speed efforts in soccer players. Sports Health, Ahead of Print: 1-12, 2024.

Background: Maximal sprinting speed (MSS) overexposure is associated with increased risk of injury. This study aimed to describe changes in sprint performance-related factors and hamstring strain injury (HSI) risk factors after a high-volume sprinting session in soccer players.

Hypothesis: A high-volume sprinting session can induce acute changes in several sprint performance-related factors (sprint time and mechanical properties) and HSI risk factors (posterior chain muscle strength, hamstring range of motion, and dynamic lumbo-pelvic control [LPC], measured as changes in anterior pelvic tilt [APT] during maximal speed sprinting).

Study design: Prospective observational case series.

Level of evidence: Level 4.

Methods: Fifteen active male amateur soccer players participated. Changes in sprint performance-related factors and HSI risk factors were examined for 72 hours after high-volume MSS efforts (H-VMSSE) using a soccer-contextualized multifactorial approach. Muscle damage proxy markers (hamstring perceived soreness and creatine kinase) were also examined.

Results: H-VMSSE induced decrements in sprint performance-related factors. Significant reductions in theoretical maximal horizontal velocity ($P < 0.01$; effect size [ES], -0.71) and performance ($P = 0.02$; ES, -0.59) were observed for 48 and 72 hours after H-VMSSE. Small but significant reductions in posterior chain muscle force-generating capacity were detected for 48 and 72 hours after H-VMSSE for the nondominant ($P < 0.03$; ES, -0.60) and dominant ($P < 0.04$; ES, -0.40) leg. Finally, players exhibited persistent small, albeit nonsignificant ($P = 0.06$; ES, 0.53), decreases in dynamic LPC (APT increases) for 72 hours after H-VMSSE.

Conclusion: H-VMSSE induced declines in both sprint performance-related factors and HSI risk factors. Sprinting can alter a player's anatomic structure by increasing APT during the maximum speed phase of the sprint.

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Clinical relevance: A soccer-contextualized multifactorial approach might allow for the regulation of MSS dosage depending on individual HSI risk factor status, thereby serving as a tailored "vaccine" for sprinting needs.

Adapting the percentage intensity method to assess accelerations and decelerations in football (soccer): Moving beyond absolute and arbitrary thresholds. Sports Biomechanics, Ahead of Print: 1-13, 2024.

Abstract - We adapted the percentage intensity approach to monitor accelerations and decelerations allowing players' individualization. Forty-two players were monitored during four microcycles via global navigation satellite system devices. Raw velocity and time data were collected to calculate acceleration and deceleration magnitudes according to specific starting speed intervals, and the efforts intensities were established as very low (75%). Linear regressions and Pearson correlation (r) analyzed the relationship between maximal efforts and starting speeds; additionally, mean paired differences compared efforts magnitudes between subsequent starting speed intervals. Most very low intensity accelerations (86%) and decelerations (79%) started from < .001) for acceleration, and r = −0.94 (p < .01) for deceleration. Maximal acceleration decreased as starting speed increases (very large effect sizes), but deceleration is less starting speed dependent (unclear to large effect sizes). This adaptation allows practitioners to individualize accelerations and decelerations classification during real-life scenarios, leading to a more precise training prescription. The very low intensity interval could be excluded to consider only relevant efforts. Maximal acceleration should be collected for each starting speed interval because accelerations are starting speed dependents.

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Alterations in hamstring properties of athletes with hamstring strain injuries can impact jump-landing performance. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-9, 2024.

Background: Hamstring strains may alter the contractile properties of the muscle and affect functional movement. Thus, this study evaluated the association between the hamstring properties and the jump-landing performance in athletes with hamstring strains.

Methods: Sixteen hamstring-injured athletes and 16 healthy controls were recruited. The mechanical properties and contractile function of the hamstring were assessed in both the injured and non-injured legs. The control group was tested only in the matched injured leg. The kinetic outcomes during squat jump (SJ) and countermovement jump (CMJ) tasks were analyzed to evaluate the jump-landing performance.

Results: The injured limb exhibited higher muscle tone (P=0.042) and stiffness (P=0.010), but lower flexibility (P=0.002) and strength (P=0.040) than the control limb. The injured limb showed a poorer jumping performance (P=0.037 for jump height) and landing performance (P=0.011 for landing force and P=0.004 for loading rate) during the CMJ task compared to the control limb. All the hamstring properties showed mild-to-moderate correlations with the jump-landing performance.

Conclusions: Impairments in the muscle properties following hamstring strain impact the jumping and landing performance, leading to degraded sports outcomes in athletes with hamstring injuries. Thus, more attention should be paid to tissue property changes following hamstring strain injuries to develop effective strategies for restoring muscle function and improving sports performance.

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American football headgear impairs visuomotor drill performance in Division I NCAA football athletes. Journal of Functional Morphology and Kinesiology, 9: 169, 2024.

Background/Objectives: Previous evidence has shown that American football headgear (e.g., facemasks, visors/eye shields) differentially impairs reaction time (RT) to visual stimuli, most notably in peripheral fields of view. However, this has only been established with stationary RT testing, which may not translate to gameplay situations that require gross motor skills. Therefore, the purpose of this study was to build upon previous findings to elucidate the effects of various American football headgear on gross motor visuomotor drill performance.

Methods: Division 1 NCAA football players (n = 16) with normal/corrected-to-normal vision participated and completed two experiments (EXP), each with differing conditions: EXP1- Varying facemask reinforcement and EXP2- Varying visor/eye shield light transmittance. In EXP1, participants completed an agility test for the following conditions: baseline/no helmet (BL), helmet + light (HL), helmet + medium (HM), and helmet + heavy (HH) face mask reinforcement. In EXP2, participants completed an agility test for the following conditions: baseline/no helmet (BL), helmet + clear visor (HCV), helmet + smoke-tinted visor (HSV), and helmet + mirrored visor (HMV). For each condition in EXP1 and EXP2, participants completed a reactive agility task using a FITLIGHT trainer system where five poles were equipped with a total of ten LED sensors and were placed in a semi-circle 1 m around a center point. Participants were asked to step and reach with their hands to hit each ten lights individually as fast as possible upon illumination. Each reactive agility test was repeated for a total of three attempts.

Results: Average reaction time was analyzed and compared between conditions and according to visual fields of interest (e.g., central vs. peripheral). Results from EXP1 showed that compared to BL, reactive agility was worsened by HL ($p = 0.030$), HM ($p = 0.034$), and HH ($p = 0.003$) conditions. No differences between facemask conditions existed for overall performance ($p > 0.05$). For EXP2, HCV ($p < 0.001$), HSV ($p < 0.001$), and HMV ($p < 0.001$) conditions resulted in worsened reactive agility performance compared to BL. No differences between visor conditions existed for overall performance ($p > 0.05$).

Conclusions: Overall, these findings suggest that American football headgear impairs reactive agility, which could result in worsened game performance and safety. Future studies investigating training strategies to overcome impairments are warranted.

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Associations between brain metabolites measured with MR spectroscopy and head impacts in high school American football athletes. Journal of Magnetic Imaging, Ahead of Print: 1-13, 2024.

Background: While changes in brain metabolites after injury have been reported, relationships between metabolite changes and head impacts are less characterized.

Purpose: To investigate alterations in neurochemistry in high school athletes as a function of head impacts, concussion, and the use of a jugular vein compression (JVC) collar.

Study type: Prospective controlled trial.

Subjects: A total of 284 male American football players, divided into JVC collar and noncollar groups; 215 included in final analysis (age = 15.9 ± 1.0 years; 114 in collar group).

Field strength/sequence: 3 Tesla/T1-weighted gradient echo, 1H point resolved spectroscopy, acquired between August and November 2018.

Assessment: Head impacts were quantified using accelerometers. Concussion was diagnosed by medical professionals for each team. Pre- to postseason differences in total N-acetylaspartate (tNAA), total choline (tCho), myo-inositol (myoI), and glutamate + glutamine (Glx), in primary motor cortex (M1) and anterior cingulate cortex (ACC), relative to total creatine (tCr), were determined.

Statistical tests: Group-wise comparisons were performed using Wilcoxon signed-rank, Friedman's, and Mann-Whitney U tests. Relationships between Δmetabolite/tCr and mean g-force were analyzed using linear regressions accounting for concussion and JVC collar. Significance was set at $P \leq 0.05$.

Results: In participants without concussion, a significant decrease in tCho/tCr (0.233 ± 1.40 × 10-3 to 0.227 ± 1.47 × 10-7) and increase in Glx/tCr (1.60 ± 8.75 × 10-3 to 1.63 ± 1.08 × 10-2) in ACC were observed pre- to postseason. The relationship between ΔtCho/tCr in M1 and ACC and mean g-force from >80 g to >140 g differed significantly between participants with and without concussion (M1 β ranged from 3.9 × 10-3 to 2.1 × 10-3; ACC β ranged from 2.7 × 10-3 to 2.1 × 10-3). Posthoc analyses revealed increased tCho/tCr in M1 was positively associated with mean g-force >100 g (β = 3.6 × 10-3) and >110 g (β = 2.9 × 10-3) in participants with concussion. Significant associations between ΔmyoI/tCr

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in ACC and mean g-force >110 g (β = -1.1 × 10-3) and >120 g (β = -1.1 × 10-3) were observed in the collar group only.

Data conclusion: Diagnosed concussion and the use of a JVC collar result in distinct neurochemical trends after repeated head impacts.

Level of evidence: 2 Technical Efficacy: Stage 3.

Bidirectional link between exercise and the gut microbiota. Exercise and Sport Sciences Reviews, 52(4): 132-144, 2024.

Exercise is well known to exert beneficial changes to the gut microbiota. An emerging area is how the gut microbiota may regulate exercise tolerance. This review will summarize the current evidence on how exercise influences gut microbial communities, with emphasis on how disruptions or depletion of an intact gut microbiota impacts exercise tolerance as well as future directions.

Key Points:

- Exercise exerts beneficial effects on the gut microbiota.
- Reducing or eliminating the gut microbiota reduces exercise capacity.
- Rationale for the reduction in exercise can be due to the alteration in the gut microbiota–skeletal muscle axis.

Circadian variation in coaches' decision-making in the National Football League's evening games. Chronobiology International: The Journal of Biological and Medical Rhythm Research, Ahead of Print: 1-8, 2024.

Abstract - The aim of this study was to explore whether National Football League (NFL) coaches show variation in their decision-making on fourth down when traveling through time zones. Data from visiting teams in games from 20 seasons (2000–2020) of the NFL were retrieved from online sources (n = 5360

games). Decision-making was measured with the percentage of offensive plays on fourth down. A factorial ANCOVA was done to verify whether travel direction had an impact on fourth downs in evening games, while controlling for the seasons. A moderation analysis was computed to verify whether the time of game moderates the relationship between longitudinal distance traveled and decisions on fourth downs. Results showed that in evening games, coaches in teams traveling westward called more offensive plays on fourth down, compared to when they traveled in any other direction. Results from the moderation analysis showed that only in evening games, further westward longitudinal degrees traveled predict more fourth downs. For the first time, this study offers insight that circadian misalignment may not only affect player performance but also influence coaching decisions in professional sports. These results beg the question whether other aspects of coaching or staff decisions show circadian variations in professional sports.

Effects of cannabidiol ingestion on thermoregulatory and inflammatory responses to treadmill exercise in the heat in recreationally active males. Medicine and Science in Sports and Exercise, Ahead of Print: 1-43, 2024.

Purpose: Exertional heat stress can induce systemic endotoxin exposure and a pro-inflammatory cascade, likely impairing thermoregulation. Cannabidiol (CBD) is protective in pre-clinical models of tissue ischaemia and inflammation. Therefore, this study examined the effects of CBD ingestion on exercise-induced thermoregulatory and inflammatory responses.

Methods: In a randomized, double-blinded study, thirteen active males (age 25 ± 5 y; peak oxygen uptake [VO2peak] 50.4 ± 3.2 mL/kg/min) ingested 298 mg CBD or placebo 105 minutes before 1 h treadmill exercise (60-65% VO2peak) in 32°C and 50% relative humidity. Core temperature, skin temperature, heart rate, subjective outcomes and sweat loss were assessed during/after exercise. Plasma osmolality, plasma volume changes and plasma markers of intestinal damage (I-FABP), monocyte activation (CD14) and inflammatory cytokine responses (IL-6, IL-8 and TNF-α) were assessed at baseline, pre-exercise, 20- and 90-min post-exercise.

Results: Core temperature (Δ 1.69 ± 0.48°C [CBD] and 1.79 ± 0.53°C [Placebo]) and I-FABP increased during exercise, with no differences between conditions ($p > 0.050$). Mean (95% CI) CD14 was 1776 (463

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to 3090) pg/mL greater 90 min post-exercise in placebo (p = 0.049). Median (interquartile range) peak IL 6 concentration was -0.8 (-1.1, -0.3) pg/mL less in CBD (p = 0.050), whilst the between conditions difference in IL-6 area under curve was -113 (-172, 27) pg/mL·270min (p = 0.054).

Conclusions: CBD did not affect thermoregulation during exertional heat stress but appeared to elicit minor immunosuppressive effects, reducing CD14 and IL-6 responses, warranting investigation in humans under more severe heat strain and other pro-inflammatory scenarios.

Imaging and surgical procedures after surface-related collegiate football injuries on artificial turf versus natural grass. The Orthopaedic Journal of Sports Medicine, 12(9): 1-10, 2024.

Background: Few longitudinal studies exist comparing medical procedures after football injuries on artificial and natural grass surfaces.

Purpose/hypothesis: The purpose of this study was to specifically compare imaging and surgical procedures after surface-related collegiate football injuries on artificial turf versus natural grass. It was hypothesized that there would be no difference in the incidence of imaging and surgical procedures, combined medical procedures, or combined substantial and severe injuries over time between these surfaces.

Study design: Cohort study; Level of evidence, 2.

Methods: A total of 39 universities across all Football Bowl Subdivision conferences were evaluated over 15 seasons (2006-2020). Playing surfaces evaluated were either a heavyweight artificial turf infill system (≥9.0 lb infill/ft2) or natural grass. Outcomes of interest included medical procedures across injury category, primary injury type, injury location, and specific procedures. Data involved multivariate analyses of variance (MANOVA) and Wilks λ criteria using general linear model procedures and were expressed as medical procedure incidence rates (IRs) per 10-game season.

Results: Overall, 2224 games were documented: 1106 (49.7%) on artificial turf and 1118 (50.3%) on natural grass. Of the 9137 total injuries reported, 4010 (44%) were surface-related cases. MANOVA

indicated significant main effects between surfaces according to imaging procedure (F 2,1738 = 4.718; P = .009), surgical procedure (F 1,539 = 5.974; P = .003), and medical diagnosis (F 2,456 = 2.643; P = .040). Post hoc analyses indicated significantly lower (P < .05) incidences on artificial turf versus natural grass, respectively, for the following outcomes: imaging procedures ordered after player-to-surface impact trauma (IR [95% CI], 1.5 [1.3-1.8] vs 2.1 [1.9-2.4]), shoe-to-surface trauma during physical contact (4.4 [4.1-4.7] vs 5.2 [4.9-5.5]), foot injuries (0.6 [0.5-0.7] vs 1.1 [0.9-1.2]), ligament sprains/tears (4.6 [4.3-4.9] vs 5.3 [5.0-5.6]), muscle strains/tears (0.2 [0.2-0.3] vs 0.7 [0.6-0.9]), surgeries performed after shoe-to-surface trauma during physical contact (1.1 [0.9-1.3] vs 1.6 [1.4-1.8]), lower body surgeries (1.8 [1.6-2.0] vs 2.3 [2.1-2.6]), and surgeries involving ligament tears (1.0 [0.9-1.2] vs 1.5 [1.3-1.7]), as well as fewer diagnoses of syndesmosis sprains/tears (0.7 [0.5-0.8] vs 1.0 [0.8-1.2]) and Lisfranc trauma (0.3 [0.2-0.4] vs 0.5 [0.4-0.7]). Trends over the 15 seasons indicated a significant rise in combined medical procedures (P = .005) and combined substantial and severe injuries (P = .0007) irrespective of surface.

Conclusion: Results indicated that collegiate football competition on heavyweight artificial turf resulted in lower incidences of imaging and surgical procedures and medical diagnoses compared with natural grass.

In-match strategies to mitigate the effect of heat on football (soccer) players' health and performance. British Journal of Sports Medicine, Ahead of Print: 1-2, 2024.

In- match strategies to mitigate the effect of heat on football (soccer) players' health and performance

- Adjust warm- up duration and structure.
- Adapt game strategy.
- Use partial- body or whole- body cold water immersion during half- time.
- Apply ice towels covering a large body surface.
- Use a combination of pre- cooling and mid- cooling techniques.
- During the 3 min cooling breaks, focus on improving hydration and thermal sensation and reducing skin temperature.
- Consider extending or incorporating more cooling breaks in extreme heat.
- Include medical personnel in decision- making.

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Is chronic ankle instability associated with contractile thickness of gluteus medius and gluteus maximus during functional movement and exercise? A systematic review and meta-analysis. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-18, 2024.

Introduction: The gluteus medius (GMED) and gluteus maximus (GMAX) play a crucial role in postural control, and postural control is impaired in individuals with chronic ankle instability (CAI). However, the association between CAI and the recruitment of these muscles remains unclear. The purpose of this study was to explore the contractile thickness of GMED and GMAX during functional movements in individuals with CAI compared to healthy controls.

Evidence acquisition: A systematic search was conducted in six databases (PubMed, Embase, Cochrane Library, Web of Science, EBSCO, and PEDro). Included studies involved the contractile thickness of GMED and/or GMAX comparing CAI with non-CAI. The extracted data were subjected to meta-analysis for observing the differences between the two. The correlation and difference in contraction between GMED and GMAX were analyzed using the Pearson Correlation Coefficient (PCC) and t-test, respectively.

Evidence synthesis: Six studies with 176 participants were found. Contractile thickness measured by ultrasound showed GMED thickness change to be lower in CAI than in controls during functional movements and exercise (WMD: -0.08; 95% CI: -0.11, -0.04; $P<0.00001$). There was no significant difference between the two groups with respect to contraction of GMAX (WMD: 0.02; 95% CI: -0.01, 0.05; $P=0.25$). The PCC and P value (t-test) between the ratio of contractile thickness of GMED and GMAX in CAI were 0.397 and 0.029 respectively, indicating activity differences.

Conclusions: CAI may be associated with weaker GMED recruitment during functional movements and exercise, but the activation of GMAX in CAI may be unaffected.

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Limitations of separating athletes into high or low risk groups based on a cut-off – A clinical commentary. International Journal of Sports Physical Therapy, 19(9): 1151-1164, 2024.

Background: Athlete injury risk assessment and management is an important, yet challenging task for sport and exercise medicine professionals. A common approach to injury risk screening is to stratify athletes into risk groups based on their performance on a test relative to a cut-off threshold. However, one potential reason for ineffective injury prevention efforts is the over-reliance on identifying these 'at-risk' groups using arbitrary cut-offs for these tests and measures. The purpose of this commentary is to discuss the conceptual and technical issues related to the use of a cut-off in both research and clinical practice.

Clinical question: How can we better assess and interpret clinical tests or measures to enable a more effective injury risk assessment in athletes?

Key results: Cut-offs typically lack strong biologic plausibility to support them; and are typically derived in a data-driven manner and thus not generalizable to other samples. When a cut-off is used in analyses, information is lost, leading to potentially misleading results and less accurate injury risk prediction. Dichotomizing a continuous variable using a cut-off should be avoided. Using continuous variables on its original scale is advantageous because information is not discarded, outcome prediction accuracy is not lost, and personalized medicine can be facilitated.

Clinical application: Researchers and clinicians are encouraged to analyze and interpret the results of tests and measures using continuous variables and avoid relying on singular cut-offs to guide decisions. Injury risk can be predicted more accurately when using continuous variables in their natural form. A more accurate risk prediction will facilitate personalized approaches to injury risk mitigation and may lead to a decline in injury rates.

Level of evidence: 5.

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Lower extremity injury rates on artificial turf versus natural grass surfaces in the National Football League during the 2021 and 2022 seasons. The Orthopaedic Journal of Sports Medicine, 12(8): 1-6, 2024.

Background: It has been argued that the use of artificial turf football fields in the National Football League (NFL) increases player injury risk compared with natural grass surfaces.

Purpose/hypothesis: The purpose of this study was to quantify the rate of lower extremity injuries occurring in NFL players on artificial turf compared with natural grass surfaces and characterize the time missed due to injury and proportion of injuries requiring surgery. It was hypothesized that lower extremity injuries requiring surgical intervention would occur at a higher rate on artificial turf than on natural grass.

Study design: Descriptive epidemiology study.

Methods: Lower extremity injury data for the 2021 and 2022 NFL seasons were obtained using publicly available records. Data collected included injury type, player position, player age, playing surface, weeks missed due to injury, and whether the patient underwent season-ending or minor surgery. Multivariable logistic regression was performed to determine the risk of season-ending surgery according to playing surface.

Results: When combining injuries for the 2021 and 2022 seasons (N = 718 injuries), the incidence rate of lower extremity injury was 1.22 injuries/game for natural grass and 1.42 injuries/game for artificial turf. The odds of a season-ending surgery were found to be significantly higher on artificial turf compared with natural grass (odds ratio = 1.60; 95% CI, 1.28-1.99; $P < .05$), while additional variables, including weather, age, position, week of injury occurrence, and history of prior injury, did not influence the odds of season-ending surgery.

Conclusion: The 2021 and 2022 NFL seasons of our analysis demonstrated a higher incidence rate of injuries on artificial turf surfaces compared with natural grass surfaces. In addition, the odds of injury requiring season-ending surgery were found to be significantly higher on artificial turf compared with natural grass.

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Medical malpractice lawsuits against team physicians in the National Football League increased between 1971 and 2020 but remain low overall. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-20, 2024.

Purpose: To examine malpractice lawsuits by National Football League (NFL) players against team physicians to understand factors involved with medical liability in the context of professional football.

Methods: A comprehensive review of publicly available legal databases, court records, and news archives was conducted in August 2023 to identify and analyze malpractice lawsuits involving NFL players, teams, and team physicians. Data on case demographics, physician subspecialty, allegations, and outcomes, including monetary amounts, were collected and analyzed. Linear regression was utilized to assess for a temporal relationship with the number of lawsuits.

Results: There were 273 physicians identified, 190 of which were orthopaedic surgeons. There were 28 lawsuits from NFL players against physicians and a significant increase in lawsuits was identified from 1971-2020 ($p=0.0086$, $R2= 0.60$). The most common anatomic locations of injury in a lawsuit were knee (n=11), spine (n=4), and foot/ankle (n=4). Other lawsuits include the result of wrongful death from heat-related injury as well as a class action lawsuit regarding pain medication prescribing practices and other individual lawsuits over anti-inflammatory use. The highest lawsuit amount filed was $180 million with at least six lawsuits filed for over $1 million, although many settlements and final verdicts were not available/sealed.

Conclusions: There was an increase in the number of medical malpractice lawsuits by professional athletes against team physicians in the NFL between 1971 and 2020, but the overall number remains low.

Clinical Relevance: Understanding the frequency and characteristics of medical malpractice lawsuits is essential for physicians, sports organizations, and policymakers to help guide appropriate healthcare delivery and risk management strategies for athletes in the NFL.

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Observation of risk for concussion following 'lowering the helmet to initiate contact' penalties in the NFL. Journal of Orthopaedic Reports, 3: 100347, 2024.

Objective: This observational study examines the associated risk for concussion following 'lowering the helmet to initiate contact' penalties in the National Football League.

Methods: Using officially published injury data on the National Football League (NFL) website, the authors reviewed NFL Players diagnosed with a concussion from 2015 to 2022 and who received a 'lowering the helmet to initiate contact' penalty from 2018 to 2020. Reported concussions were evaluated for association with 'lowering the helmet to initiate contact' penalties. The Friedman test assessed overall differences between the years. Pairwise comparisons were performed using the Wilcoxon signed-rank test with a Bonferroni correction to adjust for multiple comparisons.

Results: The average incidence of all concussions per play was 0.45% for regular-season games. From 2018 to 2020, 86 lowering the helmet to initiate contact penalties were called. Three of 86 penalties resulted in concussions, with a 3.5% incidence. The Friedman test indicated significant differences in concussion trends across the years ($\chi 2 = 16.0$, $p = 0.0003$). Pairwise analysis with the Wilcoxon signed-rank test and adjusted for multiple comparisons with a Bonferroni correction ($\alpha = 0.05/28 = 0.0018$) showed no statistically significant differences between years. There was a significant negative correlation of overall concussion incidence per year.

Conclusions: The incidence of concussion with players penalized for lowering the helmet to initiate contact is greater than the overall incidence of concussions in all plays. While there is no difference in total concussions that occur, the concussion rate has a significant negative trend between 2015 and 2022. However, it is difficult to analyze the nonimmediate effects of this penalty on players' intended play and practice, which may further decrease concussions.

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Pain medication data from the 2021 and 2022 National Football League Prescription Drug Monitoring Program. Current Sports Medicine Reports, 23(10): 348-351, 2024.

Abstract – We report 2021and 2022 prescription drug monitoring program data that include pain medication prescriptions, including over-the-counter medications, issued to NFL players by either club physicians or external medical providers and entered in the NFL electronic medical record. Of 3142 players who signed a contract with at least one NFL Club during the 2021 season, there were 14,903 prescriptions for pain medications issued to 2207 players. During the 2022 season, there were 14,880 prescription pain medications issued to 2189 players (out of a population of 3152). Notably, most pain medication prescriptions across the two seasons were for nonsteroidal anti-inflammatory agents (NSAIDs; 85.1%), whereas only 2.9% were for opioids. Despite the demanding physical nature of professional football and the need for pain management during a season, the prevalence of opioid prescriptions among NFL players appears to be similar to and, in some cases, potentially lower than previous U.S. population-based data.

Pharmacological and physiological effects of cannabidiol: A dose escalation, placebo washout study protocol. BMC Neurology, 24:340, 2024.

Background: Cannabinoids such as cannabidiol (CBD) exhibit anti-inflammatory properties and have the potential to act as a therapeutic following mild traumatic brain injury. There is limited evidence available on the pharmacological, physiological and psychological effects of escalating CBD dosages in a healthy, male, university athlete population. Furthermore, no dosing regimen for CBD is available with implications of improving physiological function. This study will develop an optimal CBD dose based on the pharmacokinetic data in contact-sport athletes. The physiological and psychological data will be correlated to the pharmacokinetic data to understand the mechanism(s) associated with an escalating CBD dose.

Methods/design: Forty participants will receive escalating doses of CBD ranging from 5 mg CBD/kg/day to 30 mg CBD/kg/day. The CBD dose is escalated every two weeks in increments of 5 mg CBD/kg/day.

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Participants will provide blood for pharmacological assessments at each of the 10 visits. Participants will complete a physiological assessment at each of the visits, including assessments of cerebral hemodynamics, blood pressure, electrocardiogram, seismocardiogram, transcranial magnetic stimulation, and salivary analysis for genomic sequencing. Finally, participants will complete a psychological assessment consisting of sleep, anxiety, and pain-related questionnaires.

Discussion: This study will develop of an optimal CBD dose based on pharmacological, physiological, and psychological properties for future use during contact sport seasons to understand if CBD can help to reduce the frequency of mild traumatic injuries and enhance recovery.

Position-specific differences in speed profiles among National Football League Scouting Combine participants. Journal of Strength and Conditioning Research, Ahead of Print: 1-8, 2024.

Abstract - This study examined the relationships between speed profiles and athletic performance measurements from the National Football League (NFL) Scouting Combine, and position-specific differences in speed profiles. Subjects included NFL Scouting Combines participants with valid scores for the 40-yard dash, as well as the 10- and 20-yard splits of the 40-yard dash (n = 2,156). Subjects were divided into the following position groups: defensive backs (DBs, n = 421), defensive linemen (DL, n = 356), linebackers (LBs, n = 261), offensive linemen (OL, n = 354), running backs (RBs, n = 217), tight ends (TEs, n = 122), quarterbacks (QBs, n = 108), and wide receivers (WRs, n = 317). Performance measures included 40-yard dash time (with 10- and 20-yard split times), bench press repetitions to failure, vertical jump height, broad jump distance, pro-agility time, and L-cone drill time. The 40-yard dash and its splits calculated vmax, τ, and amax to reflect speed profiles. One-way analysis of variances examined position-related differences. Pearson's product moment correlation coefficients examined relationships between performance measures and speed profile variables. Skill positions (DB and WR) had the greatest vmax and amax, and lowest τ, followed by big skill positions (LB, RB, TE, QB), followed by DL, then OL with the lowest vmax and amax, and highest τ. For skill and big skill positions, vmax, τ, and amax exhibited greater relationships with combine measures, whereas OL generally had the lowest relationships. The position-specific differences in this study demonstrate potential areas of strength and weakness for

certain positions. This may help guide strength and conditioning coaches desiring to improve position-specific speed and acceleration capabilities for American football players.

Practical Applications - Although the goal of athletic performance combines is to assess the potential for on-field performance for a given sport, simply assessing the raw outcome variables may yield in complete information. Specifically, in American football, maximal speed and acceleration are vital requisite performance characteristics that ultimately aid on-field performance. Therefore, calculating vmax, t, and amax from the 40 yd dash and associated splits may provide practical relevance for strength and conditioning coaches regarding sprinting and acceleration capabilities. Furthermore, the position-specific differences in this study demonstrate potential areas of strength and weakness for certain positions. Specifically, it appears that positions who are generally faster have greater acceleration profiles and achieve a higher percentage of their maximum velocity in a shorter amount of time. This may speak to the need of strength and conditioning coaches to prioritize training maximum speed, acceleration, or both depending on the results of the specific team.

Practice patterns and return to sports timing of National Football League head team physicians for ACL reconstruction. The Orthopaedic Journal of Sports Medicine, 12(10): 1-4, 2024.

Background: Previous surveys of the practice patterns for anterior cruciate ligament (ACL) reconstruction in collegiate and professional team physicians have shown bone–patellar tendon–bone (BPTB) autograft being the most common graft, with variable return-to-sports timing.

Purpose: To evaluate the current practice patterns and return-to-sports timing of National Football League (NFL) head team physicians for ACL reconstruction.

Study Design: Cross-sectional study.

Method: All 32 NFL head team orthopaedic surgeons were surveyed and asked their primary ACL graft preference, revision ACL graft preference, use of lateral extra-articular tenodesis (LET) or anterolateral ligament reconstruction (ALL), use of suture brace augmentation, return-to-sport timing, and whether

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functional testing was utilized in determining readiness of return to play. The years of experience of the physician were also evaluated.

Results: All 32 head team physicians responded, with 31 (97%) using ipsilateral BPTB autograft in the primary setting and 1 using ipsilateral quadriceps autograft. In the revision setting, 24 (75%) physicians preferred a contralateral BPTB autograft, while 5 (16%) used an ipsilateral quadriceps autograft, 1 (3%) used a contralateral quadriceps autograft, and 2 (6%) used a BPTB allograft. One respondent (3%) used suture brace augmentation "most of the time" in the primary setting and always in the revision setting. No respondents utilized an ALL or LET in the primary setting and 2 (6%) respondents always performed an LET in revisions. Five others (16%) have performed at least 1 ALL or LET in an NFL athlete. Mean return to sports was at 9 months postoperatively, with no physicians allowing players to return before 7 months. All but 1 respondent used functional testing in return-to-sports decision-making.

Conclusion: All but 1 of the 32 NFL head team physicians used BPTB autograft in primary ACL reconstruction. The majority preferred contralateral BPTB autograft in revisions, with a few considering extra-articular procedures or suture bracing in addition. On average, head team physicians allowed players to return at 9 months postoperatively, with none allowing return before 7 months. Nearly all utilized functional testing to aid in return-to-play decision-making. These homogenous findings in this high-risk and public athletic cohort aid in patient education and clinical decision-making for best treatment of ACL injuries, particularly in contact athletes.

Protocol for COACH, an evidence-based intervention for improved head impact safety in youth American football developed using a community-engaged approach. Contemporary Clinical Trials Communications, 42:101371, 2024.

Abstract - Subconcussive, repetitive head impacts sustained in collision sports may negatively affect brain health. American football practices are controlled environments amenable to intervention. Engaging community members is essential for successful development, implementation, and sustainability of viable interventions. The objective of this study is to develop and pilot test an evidence-based intervention to reduce head impact exposure in youth American football (i.e., football), using a community-engaged approach. This manuscript describes the co-design of the intervention and

associated implementation plan and the study protocol for evaluating the effectiveness and feasibility of the intervention and implementation plan. In the first part of this study, focus groups with parents and coaches, and individual interviews with organizational leaders associated with two teams at the middle school level were conducted. An anonymous survey assessing beliefs and perceptions of non-concussive head impacts was given to parents, coaches, and organizational leaders within the local youth football league. Following the football season, qualitative and quantitative data describing determinants of head acceleration events in football were shared with 12 stakeholders of coaches, league and school administrators, parents, an athletic trainer, and local university player development director. Together, we co-designed COACH (COmmunities Aligned to reduce Concussion and Head impact exposure) and implementation plan using a strategic planning approach. The preliminary effectiveness and feasibility were assessed in the second part of this study. Youth football players participating on the teams in year 1 (control teams) were fitted with mouthpiece-based head kinematic sensors which measure head acceleration events (HAEs). HAEs were collected and quantified during team activities. Preliminary effectiveness of the intervention to reduce HAEs was measured among two new teams pilot testing COACH with mouthpiece-based sensors, while simultaneously monitoring implementation of the intervention. We report our study design and evaluation, and opportunities and challenges with our approach. The results will inform a future full-scale pragmatic trial to assess the implementation and effectiveness of the intervention program.

Risk of anterior cruciate ligament tears in National Football League players by short, normal, or long rest weeks. Journal of Knee Surgery, Ahead of Print: 1-17, 2024.

Introduction: Anterior cruciate ligament (ACL) tears in National Football League (NFL) players are devastating injuries that take nearly a year to recover. Players that do return to sport have worse overall performance compared to pre-ACL tear. NFL players typically play regular season games on Sunday with the next game played on the following Sunday, allowing for 7 days between games. Deviation from the usual 7-day rest week has been proposed as a potential risk for ACL tear. The main objective of this study is to evaluate the risk of decreased rest or increased rest on ACL tear rates in NFL players.

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Methods: ACL injury data of NFL players from the 2012-2013 season and 2022-2023 season were gathered from publicly available sources. Player demographic data, position, age at time of injury, seasons played, injury mechanism, and playing surface type were recorded. Injuries were characterized as short, normal, or long week injuries. ACL tears that occurred during the preseason, postseason, or during week 1 were excluded. Descriptive statistics were calculated to report means, ranges, and percentages. Data were analyzed to determine statistically significant differences using Fisher's exact, chi-square, or one-way ANOVA tests.

Results: A total of 524 ACL tears were recorded in NFL players during the study window. 304 ACL tears were excluded and 220 fit inclusion criteria. 24 ACL tears occurred during short weeks, 68 during long weeks, and 128 during normal weeks. Players were 1.8 times more likely to tear their ACL during a long week compared to a normal week ($p<.001$), and 1.5 times more likely to tear their ACL during a short week compared to a normal week ($p=.02$).

Conclusion: The findings from this study suggest that deviation from the normal 7-day NFL week increases the risk of an ACL tear in NFL players when increasing or decreasing rest time. Further research exploring the impact short and long rest times on player injury risk should be conducted to prevent season ending injuries.

The effect of Bye weeks on injury event rates in the Canadian Football League. Clinical Journal of Sport Medicine, Ahead of Print: 1-6, 2024.

Objective: To determine the effect of bye weeks (no practices or games) on the injury event rate in the Canadian Football League (CFL).

Design: Historical (retrospective) cohort study.

Setting: CFL.

Participants: CFL athletes between 2011 and 2018.

Intervention: CFL pseudorandom assignment of bye weeks each season (2011-2013: 1; 2014-2017: 2; 2018: 3).

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Main Outcome Measures: Game injury incident rate ratio (IRR) in the week following a bye week compared with non–bye weeks. Sensitivity analyses: IRR for the 2 and 3 weeks following a bye week. We conducted exploratory analyses for combined game and practice injury events because we did not have the number of players exposed during practice.

Results: The IRR was 0.96 (0.87-1.05), suggesting no meaningful effect of a bye week on the post–bye week game injury event rate. We obtained similar results for cumulative game injury events for subsequent weeks: IRR was 1.02 (0.95-1.10) for the 2 weeks following the bye week and 1.00 (0.93-1.06) for the 3 weeks following the bye week. The results were similar with 1, 2, or 3 bye weeks. However, the combined game and practice injury event rate was increased following the bye week [IRR = 1.14 (1.05-1.23)]. These results are expected if the break period results in medical clearance for preexisting injuries; increasing pain in these locations following the bye week would now be considered new injuries instead of “exacerbations.”

Conclusions: Bye weeks do not appear to meaningfully reduce the injury event rate. Furthermore, there was no injury reduction when adding additional bye weeks to the schedule.

The effect of sport specialization on injury risk in NCAA athletes: Results from the SAFE Consortium. Clinical Journal of Sports Medicine, Ahead of Print: 1-5, 2024.

Objective: Youth athletes are beginning to specialize in a single sport more often. Previous studies in sports medicine and orthopedics have shown an association between intensity of sport specialization and incidence of injuries. This study is the first of its kind to explore the effects of early sport specialization on injury risk through a multicenter framework with a concentration on NCAA athletics.

Design: Retrospective cohort study.

Setting: SAFE Consortium.

Participants: A total of 211 collegiate athletes from the NCAA's 3 levels of competition: Division I, II, and III. Data were collected by the SAFE investigators.

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Intervention: N/A.

Main outcome measures: Participants completed a questionnaire about their demographics, sport participation, specialization status, physical injuries, recovery period, and treatment method. Specialization status was calculated with a previously published 3-point scale: low, moderate, and high. Injuries were categorized as upper extremity injuries (UEIs) and lower extremity injuries (LEIs).

Results: Highly specialized athletes were more likely to report UEIs and LEIs than low specialized athletes ($P < 0.0001$). Moderate specialization, in contrast to low specialization, was associated with a higher likelihood of LEIs ($P = 0.03$) but not UEIs ($P = 0.052$). Highly specialized athletes were more likely to report an injury of any kind.

Conclusions: The SAFE investigators found high specialization was associated with a history of UEIs and LEIs. Return to play was longer for highly specialized athletes versus low specialized athletes (112 days and 85 days, respectively). Highly specialized athletes were more likely to be from Division I and to require surgery.

The Fragility Index of risk factor for hamstring injuries. Journal of Orthopaedic & Sports Physical Therapy, 54(10): 672-678, 2024.

Objective: To determine the Fragility Index of hamstring injury risk factors, defined as the minimum number of participants who would need to change classification to make a hamstring injury risk factor statistically nonsignificant.

Design: Retrospective secondary data analysis.

Methods: Studies that investigated 1 or more risk factors for hamstring injury, and presented sufficient data to develop a 2 × 2 contingency table were included. A systematic literature search and reference screening of a recent hamstring injury systematic review were conducted to identify 78 articles. Relative risk and 95% confidence intervals were determined and then systematically recalculated by removing 1 observation from the high-risk injury count and adding it to the high-risk noninjury count. The Fragility Index for a risk factor was the number of observations required to be moved between groups until the relative risk was no longer significant.

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Results: The median Fragility Index of all hamstring injury risk factors was 3 (Q1-Q3 = 2-6). The Fragility Index for nonmodifiable risk factors was 3 (Q1-Q3 = 2-6) and 3 (Q1-Q3 = 2-5) for modifiable risk factors. Over 35% of all included hamstring injury risk factors had a Fragility Index of ≤2.

Conclusion: Most statistically significant hamstring injury risk factors are fragile associations. The interpretation of significant hamstring injury risk factors should consider a range of statistical metrics, and while the Fragility Index should never be considered in isolation, it is an intuitive measure to help assess the robustness of findings.

The importance of language in describing concussions: A qualitative analysis. PM&R: The Journal of Injury, Function, and Rehabilitation, Ahead of Print: 1-8, 2024.

Background: Concussions are mild traumatic brain injuries that are often undiagnosed due to difficulties in identifying symptoms. To minimize the negative sequelae associated with undiagnosed concussion, efforts have targeted improving concussion reporting. However, knowing more about concussions does not indicate how likely an athlete is to report their concussion. Alternatively, the attitudes and beliefs of athletes and surrounding stakeholders have shown to be a better indication of whether an athlete intends to report their concussion. Prior research has shown that athletes report concussions less often when the injury is described using language that minimizes their severity, such as when it is referred to as a "ding." This study evaluated whether describing concussions using the word "brain" was associated with individuals' underlying attitudes and beliefs about the injury's severity.

Objective: To measure the relationship between perceived concussion severity and the language used to describe concussions, specifically whether participants used the word "brain" in describing the injury.

Methods: One-on-one semi-structured telephone interviews were conducted, and a cross sectional secondary qualitative analysis was performed to assess participants' perceived concussion severity and their use of the word "brain" to describe concussions.

Design: Cross-sectional secondary qualitative analysis.

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Setting: One-on-one semistructured telephone interviews.

Participants: In 2017, 94 individuals involved in high school sports, including athletes, coaches, educators, parents of athletes, and athletic directors were recruited via convenience sampling.

Main outcome measures: Respondents' perceived severity of concussions.

Results: Individuals who used a brain phrase to describe concussion also perceived concussions as more severe ($p < .001$). Specifically, those who described concussions with maximum severity had higher odds of using brain phrases than those who described concussions as having minimum (odds ratio [OR] = 0.05, 95% confidence interval [CI] = 0.002-0.299, $p < .001$) or moderate severity (OR = 0.24, 95% CI = 0.086-0.647, $p = .003$), with the most significant relationship found among coaches.

Conclusions: These findings demonstrate the relationship between medical terminology and perceived severity of concussions. This relationship may play a role in concussion reporting behavior for coaches, athletes, and parents. Education programs using similar medical terminology may promote concussion reporting behaviors.

The management of a professional footballer recovering from a surgical repair of the proximal adductor longus tendon through a return to performance pathway – A case report. Research in Sports Medicine, Ahead of Print: 1-11, 2024.

Abstract - This case report details the management of a 23-year-old professional footballer who sustained a rupture of the proximal adductor longus (AL) tendon. Following surgical reattachment of the tendon, the player completed an eleven-phase return to performance (RTPerf) pathway designed to ensure a rapid and safe return to play (RTPlay). The pathway uses distinct phases that incorporate clinical, performance, and sport-specific criteria to guide decision-making throughout the process. This case report outlines the phases and criteria used in conjunction with shared decision-making by the interdisciplinary team (IDT) to ensure a successful RTPlay. The effectiveness of this pathway was demonstrated by the player's return to competitive play 12 weeks post-surgery.

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The relationships between reaction time scores and false start penalties of offensive linemen in the National Football League. Perceptual and Motor Skills, Ahead of Print: 1-14, 2024.

Abstract - Each year in the United States, many factors are considered in the National Football League's draft process, including scouting reports, interviews, medical tests, and measures of players' strength, speed, anthropometry, and personality. Recently, an increased emphasis has been placed on players' psychometrically measured cognitive abilities. Historically, offensive linemen have been excluded from NFL research, since they do not have individual performance statistics. In this study, we attempted to fill that research gap by investigating correlates of offensive linemen's measured vigilance/impulsivity. We obtained archival player data from 85 offensive linemen in the years from 2014-2021 prior to each of these players' NFL draft. We then used false start penalties per game (FSPG) as an index of their vigilance/impulsivity (and general success) on the field, and we related those values to Reaction Time tests through hierarchical multiple regression analyses. After controlling for draft placement, these players' higher accuracy on a Choice Reaction Time task accounted for a statistically significant percentage of the variance associated with their FSPG statistics. We discuss the practical utility for player development of these and other cognitive assessments now used in the NFL.

Training interventions for improved deceleration ability in adult team-based field sports athletes: A systematic review and meta-analysis of the literature. International Journal of Strength and Conditioning, 4(1): 1-25, 2024.

Background: Deceleration is a fundamental component of multidirectional speed by which athletes reduce the velocity of their center of mass to stop or execute changes of direction following acceleration or running at a constant velocity. Enhancing deceleration abilities is crucial for athletes as successfully executing horizontal deceleration has important implications for match outcomes in sports requiring rapid multidirectional movements. However, specific training interventions targeting deceleration are scarce. The purpose of this systematic review and meta-analysis was to examine the effects of training interventions on deceleration performance in adult team-based field and court sports athletes.

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Methods: A systematic literature search was conducted through electronic databases, SPORTdiscus, PubMed, and Web of Science from inception to February 2022, and re-run in May 2023. The search terms were related to different training interventions and kinetic, kinematic, and performance outcomes related to deceleration performance. Studies were included if they consisted of a randomized controlled trial which investigated the effects of training on deceleration-specific outcome measures in adult team-based field and court sports athletes. Risk of bias was assessed using the revised Cochrane risk-of-bias tool for randomized trials (RoB2). Post-intervention effect sizes (Hedge's *g*) were calculated between the intervention and control groups and a meta-analysis was performed using a random effects model.

Results: Twelve studies were included, with 29 deceleration-specific outcomes measured in a total of 381 participants. There was inconsistency in methodological designs, including control group types, length and type of interventions and in reported deceleration-specific outcome measures. Across all observations of deceleration performance measures there was a standardized mean difference of -0.04 (95% CI: -0.50, 0.42), favoring control groups, indicating little effect of training on deceleration performance. For secondary outcomes related to deceleration, for kinetics (SMD = -0.29, 95% CI = -0.83, 0.25, I^2 = 55%, $p < 0.01$), joint kinetics (SMD = -0.20, 95% CI = -1.01, 0.61, I^2 = 68%, $p < 0.01$) and muscle activation (SMD = -0.10, 95% CI = -0.44, 0.24, I^2 = 19%, p = 0.28) had greater effects for control groups, whereas joint kinematics (SMD = 0.07, 95% CI = -0.11, 0.24, I^2 = 0%, p = 0.77), favored the intervention groups.

Conclusion: For the deceleration-specific outcomes reported in the included studies, training was not likely to produce a performance improvement in participants compared to control groups. However, due to methodological inconsistencies between studies and observed high risk of bias, the results should be interpreted with caution. More rigorous research methods should be included in the future to address areas that may introduce potential biases. Future research should address the differences in the type, timing, frequency, and duration of the implemented training interventions for improving deceleration performance, and in the reported deceleration-specific outcome measures.

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